

Forces

Forces $\vec{F} = F_x \mathbf{i} + F_y \mathbf{j} + F_z \mathbf{k} = |\vec{F}| \vec{\lambda}$ Direction: $\vec{\lambda} = \cos \theta_x \mathbf{i} + \cos \theta_y \mathbf{j} + \cos \theta_z \mathbf{k}$

Magnitude: $F = |\vec{F}| = \sqrt{F_x^2 + F_y^2 + F_z^2}$ $\cos^2 \theta_x + \cos^2 \theta_y + \cos^2 \theta_z = 1$

$\cos \theta_x = \frac{F_x}{F}$ $\cos \theta_y = \frac{F_y}{F}$ $\cos \theta_z = \frac{F_z}{F}$

Resultant Forces

$$\vec{R} = \sum_{i=1}^n \vec{F}_i = \sum_{i=1}^n F_{i,x} \mathbf{i} + \sum_{i=1}^n F_{i,y} \mathbf{j} + \sum_{i=1}^n F_{i,z} \mathbf{k}$$

Moment

$M_O = F r \sin \theta = F d$ $\vec{M}_O = \vec{r} \times \vec{F}$ **arm of moment**

Cross-product of vectors A and B

$\vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin \theta_{AB} \vec{n}$

$\vec{P} \cdot \vec{Q} = PQ \cos \theta$

Moment of a force about an axis

$M_{OL} = \vec{\lambda} \cdot \vec{M}_O = \vec{\lambda} \cdot (\vec{r} \times \vec{F})$ $M_{OL} = \begin{vmatrix} \lambda_x & \lambda_y & \lambda_z \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix}$

Couple

$\vec{M}_O = \vec{r}_A \times \vec{F} + \vec{r}_B \times (-\vec{F})$ **Definition:** The moment produced by two equal and non-collinear opposite forces is known as couple.

$= (\vec{r}_A - \vec{r}_B) \times \vec{F} = \vec{r} \times \vec{F}$ **Direction:** normal to plane formed by the forces $\vec{F}$ and $-\vec{F}$.

The moment of a couple is a free vector, O is any point.

Free Body Diagram

Support or Connection	Reaction	Number of Unknowns
Rollers	Force with known line of action	1
Rocker	Force with known line of action	1
Frictionless surface	Force with known line of action	1
Short cable	Force with known line of action	1
Short link	Force with known line of action	1
Collar on frictionless rod	Force with known line of action	1
Frictionless pin in slot	Force with known line of action	1
Frictionless pin or hinge	Force of unknown direction	2
Rough surface	Force of unknown direction	2
Fixed support	Force and couple	3

Work $dU = \vec{F} \cdot d\vec{r}$ $dU = M d\theta$

The work of the couple is positive if M has the same sense as $d\theta$ (clockwise in this illustration) and negative if M has a sense opposite to that of the rotation.

$\delta U = \delta U' - (\delta V_e + \delta V_g) = 0$ $\delta U' = \delta V$

Motion of particle

Velocity:

$\vec{v}(t) = \dot{\vec{r}}(t) = \frac{dr}{dt} \vec{e}_r + r \frac{d\vec{e}_r}{dt}$

$= \dot{r} \vec{e}_r + r \dot{\theta} \vec{e}_\theta$

Radial velocity Transverse velocity

Acceleration:

$\vec{a}(t) = \frac{d\vec{v}}{dt}$

$= \ddot{r} \vec{e}_r + \dot{r} \dot{\theta} \vec{e}_\theta + r \ddot{\theta} \vec{e}_\theta + r \dot{\theta} \dot{\theta} \vec{e}_\theta$

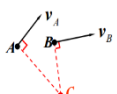
$= (\ddot{r} - r \dot{\theta}^2) \vec{e}_r + (2\dot{r} \dot{\theta} + r \ddot{\theta}) \vec{e}_\theta$

Radial acceleration Transverse acceleration

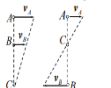
(1) Given velocity and angular rotation rate



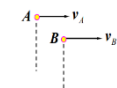
(2) Given unparallel velocities at two points



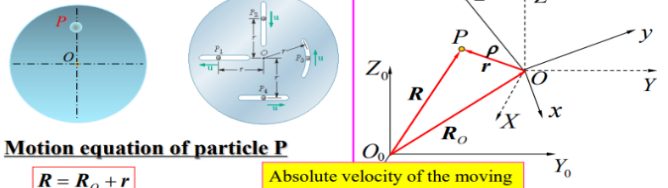
(3) Given parallel velocities at two points, normal to the connection line of two points



(4) Instantaneous translation, with $\vec{v}_A = \vec{v}_B$



Motion of a particle relative to a moving frame



Motion equation of particle P

$\vec{R} = \vec{R}_O + \vec{r}$

Velocity of particle P

$\vec{v}_P = \vec{v}_e + \vec{v}_r$

$\vec{v}_e = \vec{v}_O + \vec{\omega} \times \vec{r}$

$\vec{v}_r = \frac{d\vec{r}}{dt}$

Absolute velocity of the moving frame in the position coincide with the moving particle P

Relative velocity of particle P

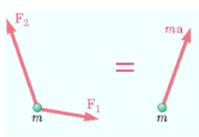
Acceleration of particle P

$\vec{a}_P = \underbrace{\vec{a}_O + \vec{\epsilon} \times \vec{r} + \vec{\omega} \times (\vec{\omega} \times \vec{r})}_{\vec{a}_e} + \underbrace{\frac{d^2 \vec{r}}{dt^2}}_{\vec{a}_r} + \underbrace{2\vec{\omega} \times \vec{v}_r}_{\vec{a}_c}$

Coriolis Acceleration

Newton's second law

$\sum \vec{F} = m\vec{a} = m \frac{d\vec{v}}{dt} = \frac{d(m\vec{v})}{dt} = \frac{d\vec{L}}{dt}$ $\sum \vec{F} = m\vec{a} \Rightarrow \sum \vec{F} = m\ddot{\vec{r}}$



Cartesian coordinate

$m\ddot{x} = F_x$

$m\ddot{y} = F_y$

$m\ddot{z} = F_z$

Tangential & normal

$m\ddot{s} = F_t$

$m \frac{\dot{s}^2}{\rho} = F_n$

$0 = F_b$

Radial & transverse

$m(\ddot{\rho} - \rho \dot{\theta}^2) = F_\rho$

$m(\rho \ddot{\theta} + 2\dot{\rho} \dot{\theta}) = F_\theta$

$m\ddot{z} = F_z$

Alternative Newton's second law

$\sum \vec{F} + \vec{F}_{inert} = 0$

Dynamic equilibrium condition

Principle of work and energy

$T_1 + V_1 = T_2 + V_2$

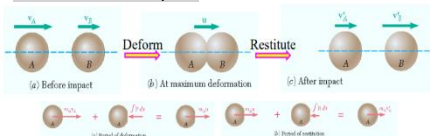
$U_{1 \rightarrow 2} = T_2 - T_1$

Total Mechanical Energy

Principle of Impulse and Momentum

$m\vec{v}_1 + \int_{t_1}^{t_2} \vec{F} dt = m\vec{v}_2$

Direct Central Impact



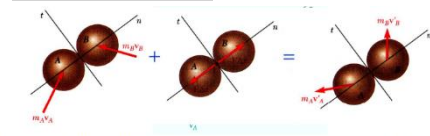
Particles system: $m_A \vec{v}_A + m_B \vec{v}_B = m_A \vec{v}'_A + m_B \vec{v}'_B$

$m_A \vec{v}_A + \int \vec{P} dt = m_A \vec{u}$ $m_A \vec{u} + \int \vec{R} dt = m_A \vec{v}'_A$

Coefficient of restitution: $e = \frac{\int R dt}{\int P dt}$

$e = \frac{u - v'_A}{v_A - u} = \frac{v'_B - u}{u - v_B} \Rightarrow e = \frac{(u - v'_A) + (v'_B - u)}{(v_A - u) + (u - v_B)} = \frac{v'_B - v'_A}{v_A - v_B}$

Oblique Central Impact



Components along t direction: $(v_A)_t = (v'_A)_t$ $(v_B)_t = (v'_B)_t$

Components along n direction:

$m_A (v_A)_n + m_B (v_B)_n = m_A (v'_A)_n + m_B (v'_B)_n$

$(v'_B)_n - (v'_A)_n = e [(v_A)_n - (v_B)_n]$

Principles

$\vec{H}_G = \vec{H}'_G$ $\frac{d\vec{H}'_G}{dt} = \frac{d\vec{H}_G}{dt} = \sum \vec{M}_G$

$T_1 + U_{1 \rightarrow 2} = T_2$

$U_{1 \rightarrow 2} = \sum \int_{s_1}^{s_2} \vec{F} \cdot d\vec{s} + \sum_{i=1}^n \int_{s_1}^{s_2} \vec{f}_i \cdot d\vec{s}$

Work done by internal forces

$\vec{L}_1 + \int_{t_1}^{t_2} \vec{F} dt = \vec{L}_2$

$(\vec{H}_O)_1 + \int_{t_1}^{t_2} \vec{M}_O dt = (\vec{H}_O)_2$

$\vec{r} = \sum_{i=1}^n \frac{1}{2} m_i (\vec{v}_G + \vec{v}'_i) \cdot (\vec{v}_G + \vec{v}'_i)$

$= \frac{1}{2} \left(\sum_{i=1}^n m_i \right) v_G^2 + \frac{1}{2} \sum_{i=1}^n m_i v_i'^2$

kinetic energy of the mass center G kinetic energy of particles w/ relative motion to G

$\sum \vec{M}_G = I_G \vec{\alpha}$

$H_x = +I_{G,x} \omega_z - I_{G,y} \omega_y - I_{G,z} \omega_x$
 $H_y = -I_{G,x} \omega_z + I_{G,z} \omega_x - I_{G,y} \omega_z$
 $H_z = -I_{G,x} \omega_y - I_{G,y} \omega_x + I_{G,z} \omega_z$

The origin of the coordinate at mass center G

$T = \frac{1}{2} m v_G^2 + T'$

$T' = \frac{1}{2} \int |\vec{\omega} \times \vec{r}_i|^2 dm_i$

$= \frac{1}{2} (I_{G,x} \omega_x^2 + I_{G,y} \omega_y^2 + I_{G,z} \omega_z^2)$

$\vec{H}_O = \vec{r}_{OG} \times m \vec{v}_G + \vec{H}_G$

$\vec{H}_G = \sum_{i=1}^n (\vec{r}'_i \times \Delta m_i \vec{v}'_i)$

